^{34}S(d,n) 1969Da12

$J^\pi=0^+$ for ^{34}S ground state.

1969Da12: a 4.95-MeV deuteron beam was produced from the 5.5-MeV Van de Graaff accelerator at University of Alberta and impinged on a CdS target enriched to 37% ^{34}S evaporated onto a 0.05-cm gold backing. Neutrons were detected by NE213 scintillators using the time-of-flight (tof) method with FWHM=1.0 ns. Measured $\sigma(E_n,\theta)$. Deduced levels, J, π, and L-transfers from Kunz-DWBA analysis of the measured $\sigma(\theta)$.

1993TeZX: ^{34}S(d,n)^{35}Cl at E(d)=25 MeV at CYRJC, Tohoku University.

^{35}Cl Levels

E(level)†	J^π†	L‡	Comments
0	3/2$^+$	2	
1220	1/2$^+$	0	
1762	5/2$^+$	2	
2645		(1)	L: weak stripping or possibly two unresolved levels.
2695			
3006	5/2$^-$	3	
3163	7/2$^-$	3	
4060			
4110			
4170			

† As given in 1969Da12.

‡ From DWBA analysis of the measured $\sigma(\theta)$ in 1969Da12.